

STOCHASTIC METHODS IN WATER RESOURCES

Unit 2: Hydrological statistics and extremes

Lecture 8: Hydrological design, Markov chains and regional frequency analysis

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Generalities

- ▶ **Planning and management** of water resources can be classified according to their 1) **use** (e.g. Hydroelectricity, water supply, recreation, etc) and their 2) **control** (e.g. drainage, flood, contamination, sedimentation, etc.).
- ▶ The **hydrologic design** for **control** is relate to extreme events of short duration (e.g. a flood). The optimal design is a balance between the cost and safety of the designed structure.
- ▶ Regarding that water is finite on earth, extreme hydrological events have **limit values** that are difficult to know and thus are quite uncertain. For **precipitation** data, the superior limit is defined by the **probable maximum precipitation (PMP)** and for **streamflow** data it is defined by the **probable maximum flood (PMF)**. This limits are used for instance in the design of excess spillways in reservoir.
- ▶ The estimation of the limits is deterministic so that the limits do not obey to a **return period**. An structure designed based on those limits might support all possible of hazards.

Generalities

- ▶ When the limits are not considered, the **hydrologic design** of the structure is based on frequency analysis, where the magnitude of the hydrological variable of interest is estimated for certain **design return period (T_D)**. For instance, for $T_D = 100$ yr the corresponding value of the design hydrological variable is expected to be exceeded in average once every 100 yr, or that the probability of the occurrence of that event in a given year is $1/T_D = 0.01$, which is quite low.
- ▶ Note that when a structure is design for a T_D , there is a **residual hazard** because there is a probability that a larger event occur during the **structure's design life**.
- ▶ To estimate the magnitudes of the design hydrological variables is needed to obtain the **frequency curves**; a methodology that has been explained before here. Note that the **frequency curves** are obtained assuming that the data sample (e.g. annual time series of extreme rainfall) comes from a **stationary process**.

Definition of design return period (T_D)

- ▶ A design of a structure can be based on the **design return period (T_D)**. Their definition is based on **design manuals** or **design codes** (e.g. manual de diseño del INVIAS o de la EAAB). Also, the T_D can be established after **risk analysis in hydrology** or from **economical analysis**, these two allow estimate an optimal T_D .
- ▶ The magnitude to the hydrological design variable is obtained from the **frequency curve** using T_D . However, it is important to follow the recommendation given by design manuals and codes regarding the type of **probability distributions** to be implemented and the methods to estimate their parameters. For instance, in USA agencies is recommended to use the **Log Pearson type III** for the analysis of maximum streamflows.

How to determine T_D ?

- ▶ According to the type of structure, and based on previous engineering experience and the recommendations given by design manuals and codes, a value or a range of values for T_D is given. For instance, for the design of highway drainage systems it is recommended $5\text{yrs} \leq T_D \leq 100\text{yrs}$, where the value depends on the traffic volume. Similarly, while a $2\text{yrs} \leq T_D \leq 25\text{yrs}$ is recommended for pluvial sewage systems in small cities, a $25\text{yrs} \leq T_D \leq 50\text{yrs}$ is recommended for large cities.
- ▶ Based on government policies, the level of hazard can dictate the range of T_D . So for instance, in coastal cities, the risk of floods can be higher so for the design of coastal levees, T_D can be up to 10000 yrs.

Definition of design return period (T_D)

How to determine T_D ?

- ▶ In Colombia, the **RAS 2000** technical regulations, establish that a $2\text{yrs} \leq T_D \leq 50\text{yrs}$ for the design of combine or pluvial sewage systems. The ultimate value of T_D will depend on the use, slope, area, etc of the system. The **INVIAS** establish the following T_D for the design of highways structures:
 - ▶ Gutters: $T_D = 5\text{yrs}$
 - ▶ Culvert $\phi = 0.9 \text{ m}$: $T_D = 10\text{yrs}$
 - ▶ Culvert $\phi > 0.9 \text{ m}$: $T_D = 20\text{yrs}$
 - ▶ Bridge $10\text{m} \leq \text{span} \leq 50\text{m}$: $T_D = 50\text{yrs}$
 - ▶ Bridge $\text{span} > 50\text{m}$: $T_D = 100\text{yrs}$

Selection of T_D based on hydrological risk

To estimate T_D is assumed a value for the **hydrological risk** (R_h). Recall that R_h is defined as the probability that the hydraulic capacity of the structure is exceeded by a hydrological hazard once during the expected life of the structure (n yrs). Accordingly

$$T_D = \frac{1}{1 - (1 - R_h)^{1/n}}$$

Definition of design return period (T_D)

Selection of T_D based on hydrological risk

- ▶ The table shows the values of T_D for different values of R_h and n . Note that the lesser R_h the higher T_D . In contrast, the larger n the higher T_D . Remember that when $T_D = n$, $R_h \approx 0.63$, which is quite high.
- ▶ For instance, in the design of a cofferdam and a diversion tunnel, it is common to adopt a streamflow design for a $T_D = 50$ yrs, which correspond to $R_h = 0.096$ or 9.6% for $n = 5$ yrs. This contrast, T_D for a design seismic earthquake, according to the design codes in Colombia, is equal to 475 yrs, which correspond to $R_h = 0.1$ and $n = 50$ yrs.

Conversely,

$$R_h = 1 - \left(1 - \frac{1}{T_D}\right)^n$$

Note that when T_D is low, R_h is total (absolute certainty of the occurrence of the event during n). For instance for the case of a pluvial sewage system, when $T_D = 2$ yrs and $n = 25$ yrs, $R_h \approx 1$. This means that with total certain the sewage system will experiment at least once an event that exceed its hydraulic capacity. There are other uncertainty associated to the structural design, construction, operation, etc.

R_h	Vida útil esperada n (años)					
	5	10	25	50	75	100
0,01	498	995	2488	4975	7463	9950
0,05	98	195	488	975	1463	1950
0,1	48	95	238	475	712	950
0,2	23	45	113	225	337	449
0,3	15	29	71	141	211	281
0,4	10	20	49	98	147	196
0,5	8	15	37	73	109	145
0,6	6	11	28	55	82	110
0,7	5	9	21	42	63	84
0,8	4	7	16	32	47	63
0,9	3	5	11	22	33	44
0,95	2	4	9	17	26	34

Hydrologic design

Definition of design return period (T_D)

Selection of T_D based on economical analysis